

QUALIFICATION OF A WELDING PROCEDURE (WPQR)

CERTIFICATE 07/202/9070/Z/0070/18/V/0331

Insp. Authority: TÜV NORD Systems GmbH & Co. KG

File No.: 8115642380

WPS No.: TPEL/WPS/ 08 -Rev.:0

WPQR-No.: TPEL / WPQR / 08 Revision: 0

Manufacturer: Thermopac Process Engineering LLP

Contract Ref.: 8115056937

Address: Plot No. B20,,MIDC Airoli, India. 400708

Requirements: Directive 2014/68/EU / DIN EN ISO 15614-1

Date of welding: 16/10/2017

RANGE OF APPROVAL	
Welding Process:	141(TIG) 141(TIG)
Weld type/execution*:	(BW) Pipe & plate BW, Branch weld $\geq 60^\circ$ & FW / (ss, nb, sl) ss ,nb,mb,sl -
Groove shape:	(Double"V") All Grooves
Parent Material / Group	(TR ISO 156081)Group 8.1 to Group 8.1) 8 ^b - 8
Parent Material Thickness: mm	(2) 1.4 to 2.6
Pipe Outside Diameter: mm	OD $\geq$ 150 in PA,PC rotated position, OD $\geq$ 500
Filler Material Type/ Designation:	S(EN ISO 14343 A-W-19 9 L)S M
Gas/ flux	Gas-EN ISO 14175 Class 11 Ar.
Type of Welding Current:	TIG-DCEN
Heat Input:	0.2 - 0.3
Welding Positions:	(PA) PA,PB
Preheat Temperature: ⁰ C min.	25
Interpass Temperature: ⁰ C	≤ 150
Post-Weld Heat-Treatment:	NA
TEMPERATURE LIMITATION:	Impact test has been conducted at (-29 ⁰ C)-with above mentioned parent and filler metal combination. Temperature restrictions according to the used parent and filler metal (see AD 2000 Mbl. W and VdTÜV-data sheet of welding consumables) have to be considered.
SCOPE EXTENSION/ LIMITATION:	Parent and filler material has to be in accordance to European Standards - filler material acc. to EN 13479, if so AD 2000 and acc. VdTÜV Mbl. 1153
SPECIAL ADVICE FOR MANUFACTURING:	See instructions of EN 1011 „Recommendations for welding of metallic materials”
EVIDENCE FOR QUALITY ASSURANCE:	For low temperature applications, testing of toughness has to be conducted additional by procedure qualification test and / or production test.
Note: Supplementary testing and repetition of procedure test are specified in AD 2000-Merkblatt HP 2/1 chapter 8 and has to documented separately. If the specified conditions are altered to any appreciable extent, a supplementary test is required. The supplementary test can be performed as a production test. In the event of the production of pressure vessels or pressure vessel components being discontinued for a period of excess of one year, procedure testing shall be repeated.	

Certified that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard indicated above. Requirements are fulfilled.

Hannover, 26.03.2016

Notified body (0045) for pressure equipment

Dipl.-Ing. Walther

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Enclosure: 1. WPS of Manufacturer No. TPEL/ WPS / 08(BW & FW) Rev.0
2. Supervision of test welding - Welding protocol Reports 2 pages.
Date-16/10/2017 Inspector
3. Results of examinations:- Satisfactory Report No. LPT Report No.1531 Dt.17
RT Report No.36 Dt.03/11/17 & MH/1712919(1) Dt.15/11/17

*Abbreviations see back page